

Final Report

Exploring Server Failure Solutions

Sponsor: Dell Technologies

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MEEN 402 - 512



"An Aggie does not lie, cheat or steal or tolerate those who do"

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Executive Summary

This report outlines the completion of the Dell Direct Liquid Cooling (DLC) senior design project, sponsored by Dell Technologies. The project's objective is to improve the reliability of DLC systems in data centers by creating a robust and scalable solution to isolate coolant leaks and mitigate damage stemming from system failures. With data centers expanding to meet growing computational demands, ensuring consistent and efficient cooling is more critical than ever.

Given the initial open-ended nature of the project scope, the team collaborated closely with Dell and course instructors to define goals and design requirements. These requirements emphasized constraints such as space, scalability, and ease of maintenance. The team then designed a prototype consisting of four modules designed to work together to isolate leaks within a rack server. The actuator module employs a rack-and-pinion mechanism powered by a servo motor to disconnect quick-disconnect (QD) valves, effectively stopping coolant flow to areas where leaks have been detected. This actuator is mounted on the elevator module, allowing it to move vertically across server levels. The prototype uses a screw elevator for its compact form factor and reliability. The water circulation module simulates coolant flow to rack servers using off-the-shelf pumps, manifolds, and hoses. An Arduino-based control module manages system operations, utilizing a limit switch to zero the elevator and allowing for manual overrides to ensure both functionality and safety. All components are housed within a modular 8020 aluminum frame for structural integrity and ease of assembly.

After completing the embodiment design phase, the team worked to assemble the prototype while overcoming challenges such as material delays and necessary design revisions. Key changes included replacing welded QD collars with shaft collars to protect ball bearings from heat damage and reinforcing the actuator housing for improved rigidity. The team also applied Finite Element Analysis (FEA) to validate structural strength and minimize deflection within the cantilever beam used to connect the actuator module and the elevator module.

The team utilized risk assessment methods such as failure modes and effects analysis (FMEA) and fault tree analysis (FTA) to identify key failure modes, such as actuator failures, elevator misalignment, and power loss. These methods led to the implementation of durable servo motors and reinforced mechanical components. The validation plan includes performance metrics such as a 10-second actuation time and an under-2mm frame deflection under a 50 lb load. The project remains within its \$2,667 prototype budget, with a projected unit cost of \$528, which could be further reduced through Dell's bulk purchasing.

Following the completion of the prototype and presentation at the Engineering Project Showcase event, the team must deliver the prototype to Dell headquarters.

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Glossary Table

Acronym	Definition
AI	Artificial Intelligence
DLC	Direct Liquid Cooling
CPU	Central Processing Unit
GPU	Graphics Processing Unit
CRAC	Computer Room Air Conditioning
CDU	Cooling Distribution Unit
SNPS	Solution Neutral Problem Statement
“L11”	Term from Dell referring to the rack level of manufacturing a server system
Manifold	Pipes used to distribute coolant to each level of the rack
Quick Disconnect Valves	Mechanical device that allows for the quick and secure connection and disconnection of fluid lines
Throttling	A technique that ensures that the flow of data being sent to a target can be digested at an acceptable rate

1. Introduction and Problem Definition

As technology continues to evolve, data servers have become vital across industries, supporting business operations, communication, and data storage. With rising computational demands, the need for effective cooling solutions has grown. Traditional air cooling methods are becoming insufficient, prompting innovations like immersion cooling—where servers are submerged in dielectric fluids—and two-phase cooling, which uses boiling liquid to remove heat from components. These approaches aim to maintain server performance and prevent failures caused by overheating.

Dell, a global leader in technology solutions, provides complete server systems and infrastructure to businesses. To support their commitment to efficient and reliable data center operations, Dell partnered with a team of five senior mechanical engineering students from Texas A&M to investigate and improve their Direct Liquid Cooling (DLC) system. The team—Adit Sanghani, Noah Sterenback, Wilson Ramsey, Danny Hoang, and Juan Lafon—was tasked with developing an innovative solution to enhance system reliability by addressing potential failure modes.

Through site visits to Dell’s headquarters and the Texas Advanced Computing Center, the team gained first-hand insight into DLC system operations and failure points. The DLC system includes a cooling distribution unit, manifolds, and cold plates, all connected by tubing with quick disconnects (QDs). While current leak-detection systems shut down the entire rack upon identifying a leak, this approach is inefficient and overly cautious when only a single server level is affected.

After background research, the team developed the following problem statement:

“Design a holistic solution capable of tracking physical metrics within the liquid cooling system and preserving component health during failures.”

This guided the team toward a solution focused on improving reliability and minimizing downtime.

The team identified key customer needs based on research and input from sponsors as summarized in Table 1 below. These needs were then rated by importance. Given the tight space in server racks and the diversity of customer systems, the solution had to be compact, modular, and easy to integrate. It also needed to be cost-effective and robust to minimize maintenance and improve uptime.

Table 1: *Customer Needs Table with Importance Scores.*

Needs	Importance score
- Compactness	0.1
- Flexibility/Ease of integration	0.15
- Cost/Scalability	0.2
- Effectiveness	0.3
- Reliability	0.2
- Repairability/upkeep/maintenance	0.05

To better understand system requirements, the team built a functional model of the DLC system. The primary function is to remove heat from servers using convective cooling. Supporting subsystems include coolant flow regulation, heat exchange (via cold plates), sensor-based monitoring, and power management. Sensors collect real-time data, enabling control systems to adjust pump speed and coolant flow dynamically, maintaining optimal temperatures and preventing overheating. The functional model create by the team is shown in Figure 1 below

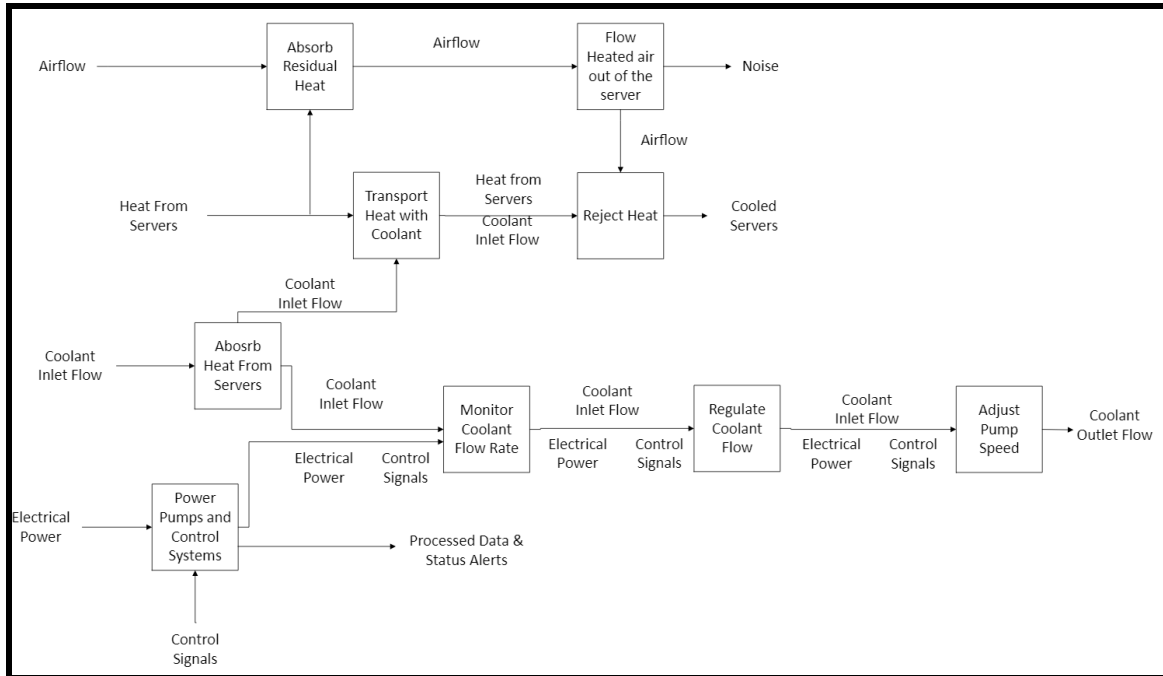


Figure 1: *DLC Core functions and component interactions*

Using this model, the team explored several design concepts before selecting a vertical-travel quick disconnect actuator. This solution isolates coolant flow at the source of a leak, preventing the need to shut down the entire rack. It also meets customer needs: it's compact, integrates easily into existing infrastructure, and improves reliability. Unlike current protocols that halt server operation upon leak detection, this system allows continued function while targeted maintenance occurs.

2. System Description

The final system is a modular, automated prototype designed to replicate the mechanical and fluid handling operations found in liquid-cooled server environments. It consists of four integrated subsystems: an actuator mechanism for bidirectional quick disconnect operation, a vertical elevator for alignment and engagement, a closed-loop water circulation system, and a control system based on the ESP32 microcontroller. These modules are mounted within a compact prototype frame built using 8020 extrusions.

The actuator is mounted on the elevator platform using a steel bracket made from bent sheet metal and reinforced with welded gussets. This platform moves vertically along aluminum rails driven by a lead screw mechanism, aligning the actuator with the quick disconnect collars connected to the water manifolds. Water flows through the system via rigid tubing connected to intake and output manifolds, forming a closed circulation loop. The entire system is governed by the control subsystem, which accepts user input to disconnect certain levels of the prototype. Table 2 shows a list of critical interfaces and the physical requirements needed for each interface. Figure 2 shows the prototype and its general assembly.

Table 2: *Critical Interface Requirements*

Interface	Requirement Specification
- Servo to Rack Gear	- Deliver 0.5 Nm of torque, reach 180 degrees of motion
- Stepper to Lead Screw	- Minimal vibration and durability
- Mounting Bracket to Elevator	- Secure attachment and proper alignment
- Mounting Bracket to Actuator	- Secure attachment and proper alignment
- Quick Disconnects to Manifolds	- Leak proof connection under operating pressure
- Quick Disconnects to Tubing	- Leak proof connection under operating pressure
- Electrical Components Connections	- Maintain connection during elevator vertical travel

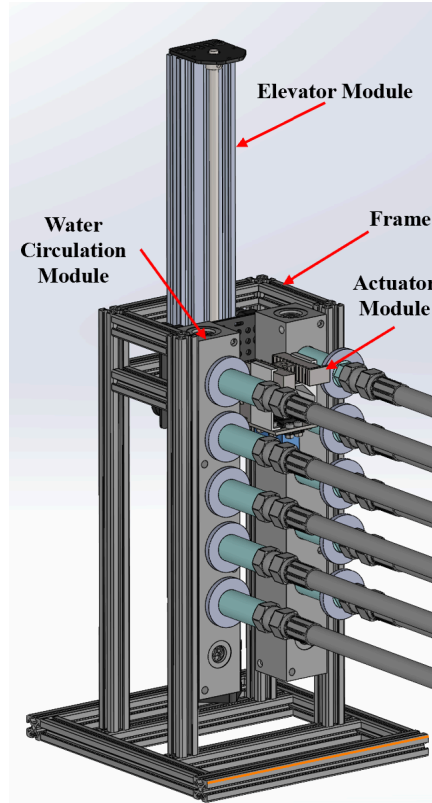


Figure 2: Prototype General Schematic

The actuator subsystem is responsible for physically engaging and disengaging the quick disconnects. A rack and pinion mechanism, enclosed in a 3D-printed housing, translates the rotary motion of a servo motor into linear motion in both directions. The rack and pinion was chosen for its mechanical simplicity and ability to operate within the limited space of the prototype. Additionally, the rack and pinion components were easily purchasable online, and an additional mechanical face interacting with the QD's was welded to the ends of the pinions. The housing was 3D-printed due to its complex geometry requiring advanced manufacturing techniques if it were created using harder materials such as steel or aluminum. Figure 3 and 4 shows general schematics of the actuator subsystem.

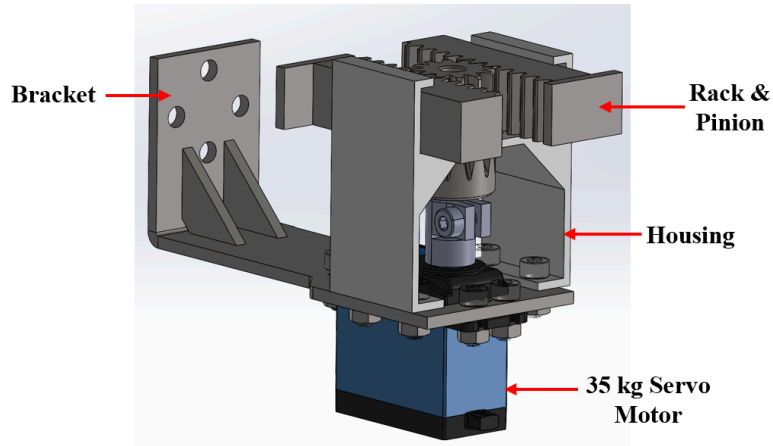


Figure 3: Actuator Subsystem

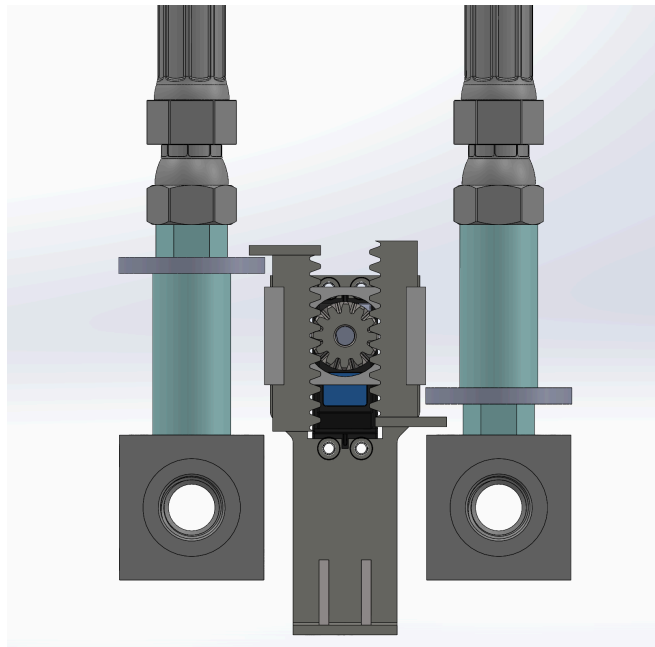


Figure 4: Actuator Subsystem Top View

Supporting the actuator, the elevator subsystem provides vertical motion to align the actuator with the manifold-mounted quick disconnects. A stepper motor drives a lead screw that translates the actuator assembly along linear guide rails. This subsystem was chosen over belt-driven design for its precision and self-locking characteristics. Upon startup, the elevator zeroes itself using a limit switch at the bottom of its travel, establishing a consistent baseline for all subsequent movements. This homing process significantly improved the repeatability of disconnect operations. Figure 5 presents the general schematic of the elevator subsystem.

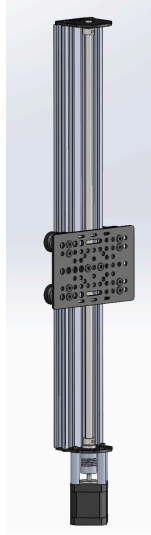


Figure 5 : Elevator Subsystem

The water circulation subsystem comprises two manifolds—one for fluid intake and the other for output—connected via rigid tubing and quick disconnects (QDs). Each QD features a detachable collar designed to enlarge the interface between the QD and the actuator, ensuring consistent and reliable engagement and disengagement. An early design proposal involved welding washers to the QDs to increase the actuation surface area; however, after consultation with staff at the Fisher Engineering Design Center, this approach was abandoned due to concerns that the heat from welding could compromise the integrity of the internal ball bearings. The selected rigid tubing offers a secure, kink-resistant flow path between connection points, enhancing reliability and repeatability during operation. To support system monitoring, a flowmeter is integrated into the intake line, allowing users to track real-time flow rate and internal pressure. Figure 6 provides a schematic overview of the water circulation subsystem.

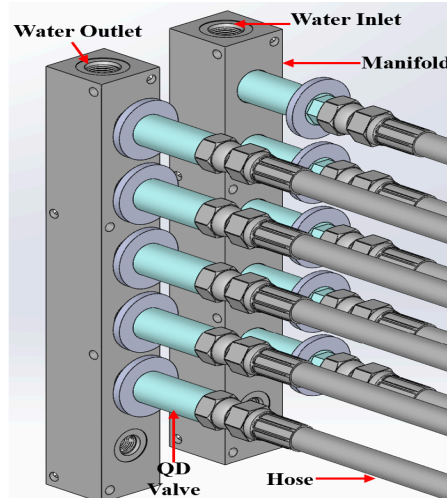


Figure 6: *Water Circulation Subsystem*

Finally, the control subsystem is centered around an ESP32 microcontroller and it interfaces with both a stepper driver and servo controller to execute motion commands based on user inputs from push buttons or toggle switches. Upon startup, the system homes the elevator using a digital limit switch, then allows the user to move to preset positions representing different operational stages. Once in position, the ESP32 commands the servo to actuate the rack and pinion, disconnecting or reconnecting the quick disconnects based on the selected mode. The ESP32 was chosen for its compact form factor. Early prototype testing consisted of calibrating the servo and step motor to ensure consistent and reliable actuation to the correct levels. Figure 7 presents the controls system.

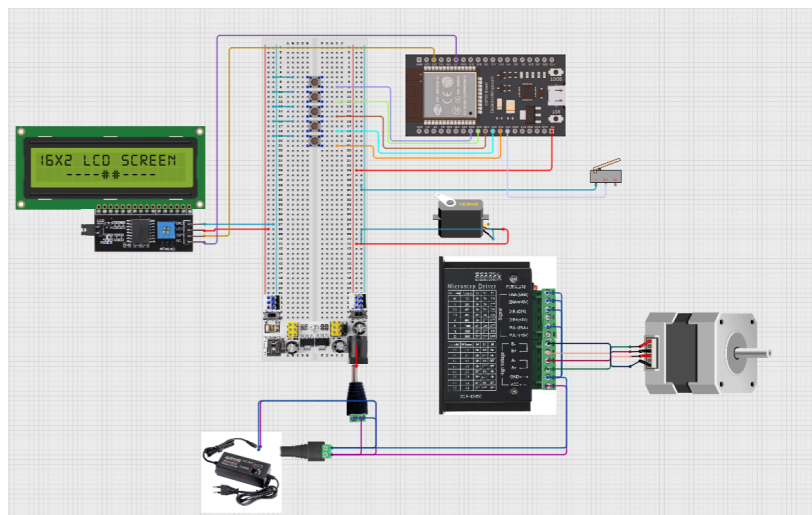


Figure 7: *Controls System Subsystem*

3. Analysis For Design

3.1.1 Embodiment Design Checklist

The embodiment design phase ensures that all critical aspects of the system have been addressed before validation and testing. This section evaluates the functional and structural integrity of the design, the effectiveness of chosen components, and their feasibility for implementation. Each aspect is discussed in relation to the key design considerations and engineering constraints outlined in the project. The checklist can be summarized as follows in Table 3.

Table 3: Embodiment Design Checklist

Aspect	Details
Functionality	Rack-and-pinion system isolates coolant flow, preventing leaks.
Working Principles	Uses vertical travel module for precise node level locating, rack and pinion push/pull for dual qd disconnection
Materials	Hardened steel for rack, pinion, cantilever beam, and housing in the future
Energy & Efficiency	Low power consumption needed (24V and 2A); servo motor outputs 3.5x required torque
Safety	Auto-reset to zero position to remain out of way in case of failure, low shock potential (24V, 2A)
Usability	Simple interface, minimal training required, user manual included
Production & Assembly	Pre-sourced parts, bolt connections for easy setup & maintenance.
Lifecycle & Cost	Long-term durability (should last longer than lifetime of servers), cost-effective with minimal maintenance

Functionality

The current design successfully meets the primary customer requirements by providing a fail-safe mechanism for isolating coolant flow in Dell's Direct Liquid Cooling (DLC) system. The product architecture consists of a rack-and-pinion mechanism, quick disconnect (QD) valves, and a belt-pulley system for vertical travel. This architecture ensures that the system operates efficiently within existing server racks while maintaining modularity for future scalability. The stipulated function of the system—to act as mitigation against liquid coolant

leaks—has been successfully incorporated. The rack-and-pinion system effectively actuates the QD valves, preventing potential water damage in case of a failure. Additionally, auxiliary functions could include a manual override mechanism for redundancy and a visual indicator system to confirm proper engagement of the actuation system in future design iterations.

Working Principles and Form Solutions

The design relies on mechanical actuation through a rack-and-pinion mechanism to engage the QD collars, ensuring a secure shutoff of the affected server node level. The form solution effectively converts rotary motion into linear motion, allowing for precise actuation of the disconnect valves. Potential noise factors are minimal since actuation occurs in discrete failure scenarios rather than continuous operation. Byproducts include negligible wear on the pinion system due to repeated actuation cycles, which has been accounted for by selecting hardened steel for critical contact points. One byproduct of this system is the potential for small amounts of coolant fluid to escape the system. This byproduct would not be more than what is already experienced in Dell's current systems.

Layout, Geometry, and Materials

The system design incorporates modular T-slotted aluminum framing for easy assembly and adaptability. The cantilever arm and rack-and-pinion housing underwent modifications to improve rigidity, with thickness increased from 2 mm to 6 mm at key load-bearing points and infill increased from 50% to 80% in 3D-printed components. These changes significantly reduced deflection while maintaining a lightweight structure.

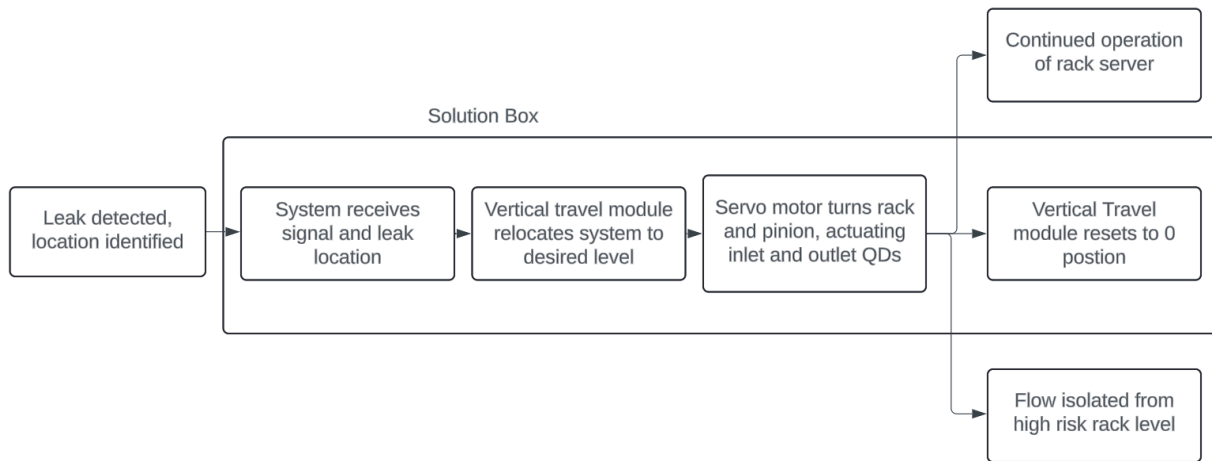


Figure 8, Functional Model.

The chosen materials ensure robustness, durability, and efficient strength-to-mass ratio, crucial for minimizing stress concentrations while maintaining stability. The aluminum frame and steel mechanical components provide high impact resistance and prevent resonance effects during operation. Figure 8 from above illustrates the system's functional flow that was used to determine the layout, material selection, and geometry of the prototype.

Energy and Kinematics

The system is designed for low energy consumption, relying solely on mechanical movement driven by servo and stepper motors. Power usage remains negligible because the actuation occurs only during failure events. The power draw is relatively small compared to the amount the servers already require. The elevator mechanism's linear motion ensures smooth engagement with the QD collars, providing a controlled shutoff with minimal resistance. The power will be tied into servers already in use, eliminating the need for an extra power supply.

According to static analysis, a force of 20 Newtons is required to push or pull off the quick disconnects. The servo motor outputs 35 kg-cm of torque, but based on our calculations, we only require 10 kg-cm of torque. Therefore, the current servo motor outputs 3.5 times the required torque while requiring small amounts of power. This system is extremely efficient because the only energy loss will be due to frictional elements. All motion is directly translated from rotational to linear motion.

Safety Considerations

The system is designed to return to a zero position when not in use, keeping it out of the way to prevent interference with normal operations. That way the actuator can't affect any pre-existing components already there if it malfunctions. Electrical components operate at a maximum of 24 volts and 2 amps, ensuring low power requirements and reducing potential electrical hazards. Additionally, all materials are corrosion-resistant, ensuring longevity in high-moisture environments. The system will also have its own separate manual to allow for operators to better understand how it works.

Ergonomics

The design avoids unnecessary complexity, ensuring that all interactions are intuitive and require minimal training. Additionally, the system is structured to be compact and unobtrusive, reducing spatial constraints in data center environments. User manuals will be provided to reduce stress from confusion on operators.

Production and Quality Control

The system's production process has been optimized for cost-efficiency and manufacturability, with primary components sourced from established vendors rather than manufactured. The waterjet cutting process ensures precision and repeatability in manufacturing cantilever arms, while preassembled quick disconnect collars reduce fabrication complexity. Quality control measures include tolerance checks on all machined components and verification of actuation reliability before deployment.

Assembly and Maintenance

The modular nature of the system ensures ease of assembly, with all major components attached via bolted connections rather than permanent fixtures. The use of T-slotted gussets allows for adjustable mounting, simplifying alignment with the manifold. Maintenance procedures are straightforward, allowing for quick inspections and part replacements without extensive downtime.

Transport Considerations

The system is designed for easy transport, with all components capable of being disassembled and reassembled with minimal effort. Packaging considerations include protective foam inserts to prevent damage during shipment. Additionally, all components are lightweight, minimizing handling risks. The housing containing the rack and pinion could be protected with foam and with a few simple wire connections the system can be ready for operation.

Lifecycle and Cost Considerations

The system is designed for long-term use, with minimal maintenance required beyond periodic inspections. A simple running of the system once a month would be sufficient in testing if components need repair. The decision to purchase off-the-shelf parts and prebuilt elevator components reduced both manufacturing complexity and cost. While cost is not the primary constraint, minimizing expenses where possible remains beneficial. Dell's ability to source materials at lower costs further enhances feasibility for large-scale implementation. The design prioritizes reliability over cost, recognizing that avoiding server damage from coolant leaks is the key financial advantage.

Scheduling and Delivery

The project remains on track to meet expected delivery deadlines, with all major components sourced and manufactured within the anticipated timeline. Potential design modifications to reduce cycle time include pre-assembling subsystems before final installation and refining alignment procedures to decrease setup time.

The current embodiment design successfully meets all functional, structural, and operational requirements. By optimizing component and material selection, manufacturing methods, and modularity, the system remains cost-effective, reliable, and adaptable. The design is now prepared for validation testing, where further refinements may be made based on real-world performance evaluations.

3.1.2 FMCA

Risk analysis is a crucial component of the design process, where the project team systematically identifies and evaluates potential risks that could impact the project's success. This process involves assessing the likelihood and severity of each risk, enabling the team to prioritize those that pose the greatest threat. Preventative measures and mitigation strategies will be developed and implemented to reduce the chances of these risks occurring or minimize their impact should they arise. By conducting a thorough risk analysis, the design team can proactively address potential challenges and uncertainties, leading to more informed decision-making and a smoother project cycle. This foresight allows for better planning, resource allocation, and contingency strategies, ultimately ensuring that the project stays on track and meets its goals with greater efficiency and safety.

The team conducted a Failure Modes and Effects Analysis (FMEA) to identify potential risks associated with the prototype design, assess their likelihood of occurrence, evaluate their severity on the project's outcomes, and determine how easily these failures could be detected after they occurred. Five key components within the prototype design were identified as having potential failure modes: the QD actuator, elevator, QD collar, electrical control system, and frame support. Once the potential failure modes were identified, each was assigned ratings across three categories: severity, occurrence, and detection. Severity indicated the level of impact a failure would have on the overall success of the project, with higher scores reflecting more detrimental consequences. Occurrence represented the likelihood of the failure mode happening, with higher scores indicating a greater chance of occurrence. Detection scores assessed how difficult it would be to identify the failure after it happened, with higher scores indicating that the failure would be harder to detect.

Once each failure mode was assigned a score in the three categories, the values were multiplied together to obtain the Risk Priority Number (RPN). The RPN serves as a measure of how critical a failure mode is to the success of the project. Failure modes with higher RPN values were considered more critical and prioritized for corrective action. These high-RPN failure modes received special attention to ensure that mitigation strategies and preventative measures could be implemented, reducing the overall risk to the project's success.

After calculating the failure modes and their respective RPNs, recommended courses of action were developed for each failure mode to provide clear guidance in case the failure occurs during the project. Furthermore, any failures that have occurred and been addressed are retroactively included in the FMEA study to prevent recurrence. An example of when the FMEA study gave the team insight on how to address potential failure modes was when the motor malfunction failure mode for the actuator encouraged the team to buy a servo motor with a factor of safety over 3 to ensure it has the required power to actuate QDs. The figure 4 shows an example of a component with potential failure modes identified, RPN number calculated, and recommended and taken actions. A finalized version of the FMEA study is provided in the appendix.

Table 4: Failure Modes Effect Analysis Study for Actuator

Part # and Functions	Potential Failure Mode	Assessment							Recommended Actions		Action Results
		Potential Effect(s) of Failure	Severity (S)	Potential Causes and Mechanisms of Failure	Occurrence (O)	Current Design Controls Test	Detection (D)	RPN	Description of Action	Responsibility & Target Completion Date	Actions Taken
Actuator									Develop a semi-annual maintenance schedule to lubricate and replace actuator components		
	Gear Slippage	no QD actuation	6	Missalignment between	2	Run the actuator 100 times to check for this particular failure.	5	60			
	Motor Malfunction	no QD actuation	6	Excessive load or po	4	Motor is rated to generate 3.25 times the required torque.	1	24	Include hard locks in the code to prevent the motor from running over the stipulaeted 312 steps/sec.		Purchased higher motor with 3.25 FOS
	Misalignment	damage componenets during vertical travel	6	Motor Doesn't Return	5	Proper greasing of all parts that come in contact with moving parts.	1	30	Make sure that ISO 1328-1 standards are followed for lubrication and alignment of gears.		

3.1.3 FTA

In addition to conducting a Failure Modes and Effects Analysis (FMEA), the design team also employed Fault Tree Analysis (FTA) to further assess potential risks and failure modes within the project. The FTA diagram begins with a top-level primary failure event that the team seeks to understand and prevent, followed by a branching structure that connects to underlying events that could contribute to the top-level failure. The diagram incorporates AND/OR logic gates to help illustrate how different events interact and lead to the failure. This tool helped the team identify critical failure modes, develop effective mitigation strategies, and enhance the overall system reliability. An example of this analysis was carried out on the elevator component shown in the figure below. While the fault tree diagram is valuable for understanding the

relationships between different failure modes, it is especially useful when analyzing complex systems with many interdependent parts. Given the simplicity of the prototype, the fault tree diagrams do not include AND gates, and none of the events have associated probabilities, which further simplifies the analysis. This also means the probability of failure cannot be calculated. Despite the limitations of the FTA diagrams for our prototypes, it was still valuable to visualize what failure conditions needed to be met for certain failure modes to occur. This, coupled with the FMEA allowed the team to understand the risks and potential pitfalls of the prototype and create plans and designs to mitigate the likelihood of failure. *Figure 9* shows a Fault Tree Analysis diagram for the elevator failure scenario.

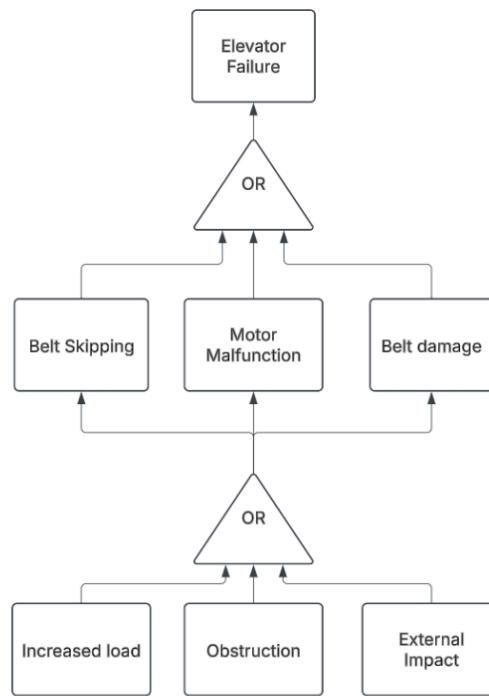


Figure 9: Fault Tree Analysis Diagram for Elevator Failure

3.1.4 Standards and Codes

The successful implementation of our project relies heavily on adherence to engineering standards, particularly **ASME Y14.5-2018** for Geometric Dimensioning and Tolerancing (GD&T). Precise tolerancing is critical for our system, especially in the rack-and-pinion mechanism, servo mounting features, and the elevator platform interface. Without tight control

of critical dimensions, components risk misalignment, backlash, or interference—failures that would compromise the functionality of the entire assembly. GD&T was used to define datums, parallelism, and positional tolerances on our detailed drawings, enabling both proper assembly and potential automation in manufacturing.

Beyond tolerancing, we also referenced several additional standards to guide design and ensure downstream compatibility:

- **SAE J1739 (2021)** – Used to assess risk via Failure Modes & Effects Analysis (FMEA) in key subsystems.
- **ISO 4162 (2012)** – Standardized metric fasteners across the system for compatibility and sourcing.
- **ASME B18.3 (2012)** – Defined specs for socket head cap screws in areas requiring fine assembly.
- **ASME B1.20.1 (2018) & ASME B31.3 (2024)** – Considered for future fluid-handling integration.
- **ASME P30.1 (2024)** – Referenced for load handling safety during installation.
- **ASHRAE 90.4 (2022)** – Informed potential energy efficiency concerns in server-environment applications.
- **IP53 (IEC 60529)** – Used as a target environmental protection level for housing components.

Together, these standards informed both the prototype and considerations for a scalable, manufacturable final product.

3.2 Design For Manufacturing and Design For Assembly

Our design process incorporated several Design for Manufacturing (DFM) and Design for Assembly (DFA) strategies to streamline prototype development and inform future production considerations.

Design for Manufacturing (DFM)

We made deliberate design decisions to reduce complexity and cost in manufacturing. One example is the rack and pinion system, which replaced an earlier concept involving two separate mechanisms. By integrating two racks and a single pinion, we significantly reduced the part count and simplified the system.

The cantilever beam was custom-designed but straightforward to fabricate using sheet metal. Our process used waterjet cutting and a single bend operation, followed by welded supports to reduce deflection. If a thicker sheet metal is used, welding could be eliminated entirely—simplifying the process further.

The housing posed the most significant manufacturing challenge. For the prototype, we 3D printed this part, but its geometric features make it a strong candidate for injection molding in production, where mold costs are justifiable.

Our elevator track and elevator platform are made from standard, off-the-shelf components. While the track required no modifications, the elevator platform includes many small parts, making it difficult to automate. A future redesign could dramatically improve manufacturability and reduce complexity.

Design for Assembly (DFA)

We standardized fasteners across the system, which simplifies both manual and automated assembly. Most parts are intuitive to put together, with clear interfaces, and the system can be easily mounted to a rack cabinet. We also ensured that it's straightforward to disassemble individual parts for replacement or recycling.

That said, we identified several opportunities for improving assembly. Some components—like the elevator platform—could benefit from a redesign to reduce the number of small parts and tedious sub-assemblies. We also noted that attaching the cantilever beam to the platform is time-consuming and could be optimized with a better joining strategy.

In future iterations, we recommend incorporating asymmetry into certain parts to prevent incorrect orientation during assembly, and simplifying servo mounting to reduce labor time and potential for error.

3.3 Design Validation

Design validation to be covered in detail. For projects utilizing a prototype, discuss the

manufacturing, assembly, and testing procedures used to validate. For projects using either an FEA or CFD, discuss assumptions used. In all cases, provide a thorough discussion of your results and how they calibrate versus your initial projections.

3.4 Comparison of the design to the requirements

Table 5: Design Comparison Table

Function	Performance Required	Design Feature which Accomplished	How Satisfactory?
Actuation Force disconnects QDs	1.25 N-m torque required	35 kg Servo Motor with rack and pinion	Servo generates 3.4 N-m of torque and proven to successfully disconnect QDs as intended
Precise Lead Screw Alignment	Aligned with center of QDs (~6 mm)	Lead Screw with mounted Cantilever Beam	Stepper motor always takes the same amount of steps and positions lead screw within 6 mm of desired location
Timely Actuation	Total actuation time <15 s	Control System	Generated times <15 s for level 5 disconnect, also did not run stepper motor at maximum speed
Node-level Flow Isolation	No visual fluid leaking out of System	Closed Loop Fluid System	Water Testing Revealed no leaking fluid and decreased flow rates after disconnect, indicating flow isolation of that level
Repeatability	System Executes 50 times	Entire Assembly and Controls	Completed 50 consecutive dry runs at varying levels successfully

Table 5 above compares the original system requirements to the final design performance. The main modeling element that was utilized in this project was Finite Element Analysis (FEA). FEA helped predict deflections in the actuator bracket and cantilever beam, leading to material selections and design changes that minimized deformation during operation. Physical testing confirmed that real-world deflections matched the FEA results.

Torque requirements for the quick disconnect valves were modeled using a force testing device pressed against the quick disconnects until they actuate. This showed that about 1.25 N-m of

torque was needed to disengage a valve. The team selected a servo motor capable of producing 3.4 N-m of torque, providing a comfortable safety margin. Testing proved that the servo successfully disconnected valves under normal operating conditions.

To meet the alignment requirement, the stepper motor-driven lead screw was tested for accuracy. Testing showed that it consistently positioned within 6 mm of the target location, even without running at full speed. The control system was also validated through timing tests, which showed that disconnects could occur in under 15 seconds as required.

Finally, water circulation tests confirmed that the system could isolate individual server levels without leaks, and the assembly completed 50 consecutive dry runs at different levels without failure. These results showed that the system reliably met or exceeded all key functional requirements.

3.5 Cost Accounting and Cost Model

This section provides a summary of the project's total expenses and a cost model for future implementation. It reviews the final budget compared to the original estimate, highlights the primary sources of expenses, and presents an estimated per-unit cost for critical hardware. A full individualized item breakdown, including unit prices and quantities, is available in the appendix for detailed reference.

Table 6: *Final Expense Breakdown by Vendor*

Vendor	Expenses
McMaster	\$1,278.12
Amazon	\$823.60
OpenBuilds	\$312.60
ServoCity	\$60.81
Grainger	\$53.60
Travel	\$282.74
FEDC Labor	\$438.00
Subtotal	\$3,249.47

Table 6 above shows the final expenses associated with each vendor. McMaster and Amazon accounted for most of the hardware and control system components, while OpenBuilds provided the elevator and lead screw assemblies. ServoCity and Grainger supplied smaller mechanical parts. Labor costs are shown separately to capture waterjet cutting and other machining services.

Overall, the total project expenses were \$3,249.47, slightly exceeding the original \$3,000 budget due to an additional trip to Austin, Texas in order to deliver the final prototype to the sponsors.

Table 7: Future Implementation Cost Model

Description	Item Total
Elevator/Lead Screw with Stepper Motor	\$190.00
Pinion Gear	\$14.14
Rack	\$12.50
Servo Motor for Rack-Pinion	\$28.99
Step Motor Connection	\$10.89
D-Shaft	\$2.09
Shaft Coupler	\$9.99
Fasteners	\$12.00
Limit Switch	\$6.00
Housing Manufacturing	\$65.00
Cantilever Manufacturing	\$55.00
Subtotal	\$406.60

Table 7 above presents the future implementation cost model, estimating the cost to scale the design for future production. It highlights the core components needed, including the elevator system, rack and pinion hardware, motors, fasteners, and limit switches. Manufacturing costs for the cantilever and housing were also included. The estimated unit cost for the critical hardware came out to \$406.60, although this model does not account for additional control system integration costs.

4. Broader Impacts of Design

As the project concludes, it is important to consider the broader impacts of the design beyond its immediate application. In addition to improving reliability in data centers, the system has wider implications across industry, safety, automation, and sustainability. This section also discusses lifecycle design considerations and potential intellectual property opportunities associated with the final product.

The quick disconnect actuator system developed in this project has potential applications across a wide range of industries. While originally designed for server rack cooling systems, the core

concept can be adapted for manufacturing, automotive, aerospace, HVAC, and medical systems, anywhere that fluid quick disconnects are commonly used. The design directly improves data center protection by mitigating the risks of coolant leaks, reducing hardware damage, and minimizing costly downtime.

The system also introduces significant cost savings. Downtime for a single 42U server rack is estimated to cost about \$80,000 for just four hours, and the losses continue to rise the longer the system is offline. By enabling faster leak isolation and reducing the need for complete shutdowns, the design has the potential to save thousands to hundreds of thousands of dollars in equipment protection and service continuity. From a safety perspective, the ability to remotely or automatically disconnect fluid lines reduces the need for human intervention, limiting exposure to hazardous fluids and improving workplace safety standards.

In terms of future readiness, the design supports the ongoing shift toward automation by enabling intelligent leak response without the need for constant human monitoring. This allows maintenance personnel to focus on more critical tasks and improves overall operational efficiency. Additionally, the design contributes to sustainability efforts by reducing electronic waste associated with hardware failures. It relies heavily on off-the-shelf and reusable parts, making repair and replacement easier while minimizing the environmental impact.

4.1 Lifecycle Design

Lifecycle considerations played an important role in both the prototype and the envisioned production version of the system. Material selection focused on durability and environmental responsibility. The design uses corrosion resistant metals like steel and aluminum, which are recyclable and extend the operational life of the system. Wherever possible, fasteners were chosen instead of welded joints. This decision reduces manufacturing complexity, allows easier repairs, and supports material separation at the end of life.

During usage and maintenance, the modular structure of the system allows for quick part replacement with minimal downtime. Standardized components simplify future repairs, making the system easier to service without specialized tools. The use of a low voltage control system also reduces overall energy consumption during operation, lowering the environmental footprint over time.

At the end of the system's service life, the design supports easy disassembly and recycling. Electronic components can be removed separately, minimizing electronic waste and making responsible disposal more practical. The system can be refurbished through simple component swaps or, if still functional, relocated to another rack even after the original server it protected is retired. Potential future programs could also include manufacturer take back options or

refurbishment services, extending the system's lifecycle and further reducing its environmental impact.

4.2 Intellectual Property

Early in our conversations with our Dell sponsors this semester, they mentioned the possibility of submitting for intellectual property in the form of a patent for our project. As the semester progressed, we submitted documents to the Dell team and met with one of their lawyers to give them a better idea of the project, and parts of the system that is worth including in the patent request. As of now, they are still processing it all internally, but there is a high likelihood that it will become patent pending in the upcoming months. The specific IP is protecting a node-level server isolation solution that utilizes a vertical travel quick disconnect actuation device. It is broken into two modules: The vertical travel module and the actuation module. It receives an outside indicator of a leak from the preexisting leak detectors, and sends a request to move the actuation module via the vertical travel module to the correct level from the leak indicator. The system then actuates the leakless quick disconnect, stopping the flow to the individual server. The patent contains embodiments for different methods of vertical travel and actuation than the lead screw elevator and rack and pinion system we utilized for our prototype. This includes belt, rack and pinion, scissor lift, and other vertical travel mechanisms as well as linear solenoid, cam and follower, and other actuation mechanisms. Overall, we are excited to potentially be named on a Dell patent, and that it contains multiple embodiments of our design.

4.3 Liability Considerations

A lot of liability for both the prototype and real-life integration of our solution stems from human error. If that is the case of product misuse, misinstallation, or manufacturing errors, the result is a problem that can cause Dell potential problems. To mitigate these issues, there are a number of steps that can be taken to remove the potential liability of the product. The first is creating proper user instructions and teaching user safety for the product. This means proper documentation, user manuals, and troubleshooting guides are needed to ensure no misuse. It also means ensuring the product falls in line with regulatory requirements for data server technology. Dell also needs to put in place procedures for product recalls in the case of a manufacturing defect. They most likely have something similar in place, so they need to ensure that the node-level isolation system is included in that plan. They need to make it clear the responsibility of all of the parties involved, such as manufacturers and suppliers. Dell also needs to consider liability insurance in the case of unforeseen issues that may arise in order to protect them as a company. In sum, thinking ahead of unforeseen circumstances and liabilities, and planning for them will provide positive results for the implementation of our system for Dell.

4.4 Ethical Considerations

The nature of our project and product solution does not lend itself to a lot of ethical considerations. A possible ethical dilemma would be in the case of human replacement. Anytime a task is automated, it replaces the role that a human would be doing instead. In some cases, this can cause people to lose their jobs due to the reduced need for labor in a facility. In our case, it would be more likely for it to just ease the burden of work for the preexisting staff, and not cause job replacement. It likely means that they will have more time to fix other issues instead of tending to the request to remove a leaking server. There is also still the need to fix and replace the server that was having issues, which still requires human resources. Another possible ethical consideration is the possibility of injury. Anytime there is a mechanical assembly with moving parts, there is the potential for human injury. Some possibilities would be a finger or hair stuck in either the elevator or the actuator. To mitigate this issue, installing proper safety equipment would be paramount. Some possible solutions would be light curtains installed on the back of the server that would ensure the system would not operate if there is something obstructing the space. Another possibility is installing sensors or cameras to detect people or obstructions in the way and stop the system. Overall, there are very few ethical considerations, but the few that there are can be overcome with preventive planning.

5. Summary

During our project, we utilized many tools to ensure that we stayed on track and completed the project on time. Beyond this, we also needed to consider the technology development required, limitations of our solution, as well as the future work required. These tools and analyses allow for a better understanding of our project as a whole, and give more insights into the process that was involved in creating our prototype and how to successfully apply it to a real application.

5.1 Work Breakdown Structure

To showcase the work packages needed to implement the design, we created a Work Breakdown Structure (WBS). Shown below in **Figure 10**.

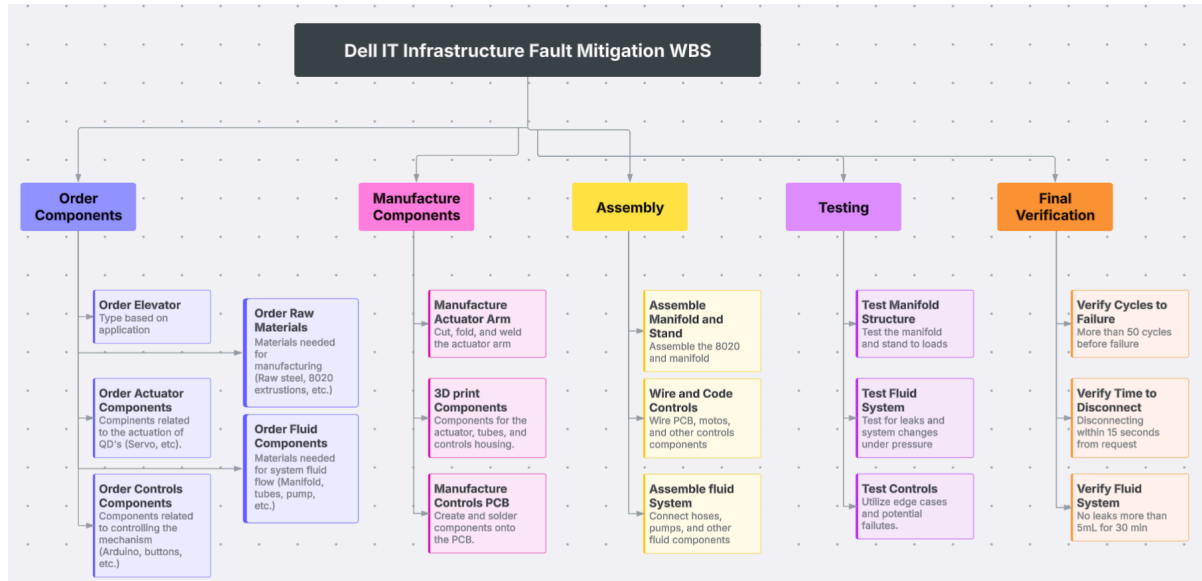


Figure 10: Work Breakdown Structure for Solution Implementation.

The WBS focuses on the tasks needed to implement the design, not only for our prototype, but also gives insight into how it would look for the real implementation as well. The first main step is to order the components needed for the manufacturing and assembly. This included a lot of the off-the-shelf components as well as the raw materials that are later processed. The next step is manufacturing, where the parts that can not be bought off the shelf are created. After that, assembly is required, combining the ordered components with the manufactured parts to create a system that works like our prototype. Next, testing is required to ensure that the assembly was done properly and that there were no manufacturing defects. Along with this, validation is required to ensure that it meets the operational requirements for a real implementation. Overall, this work breakdown structure allowed us to work through the process to implement the design.

5.2 Gantt Chart and Project Plan

Shown below in **Figure 11**, near the end of the previous semester, our team prepared a Gantt chart to track our progress and quickly identify any deviations from our plan, allowing us to develop mitigation strategies as needed. Over the course of this semester, we did go off track a few times due to administrative issues like the Mechanical Engineering Business Office losing our purchase orders, or vendors delivering wrong part numbers. However, when we initially developed the gantt chart, we made sure to leave enough buffer time between activities on the critical path. This made sure we were always on track. Also, making sure we ordered extra materials helped mitigate scheduling errors.

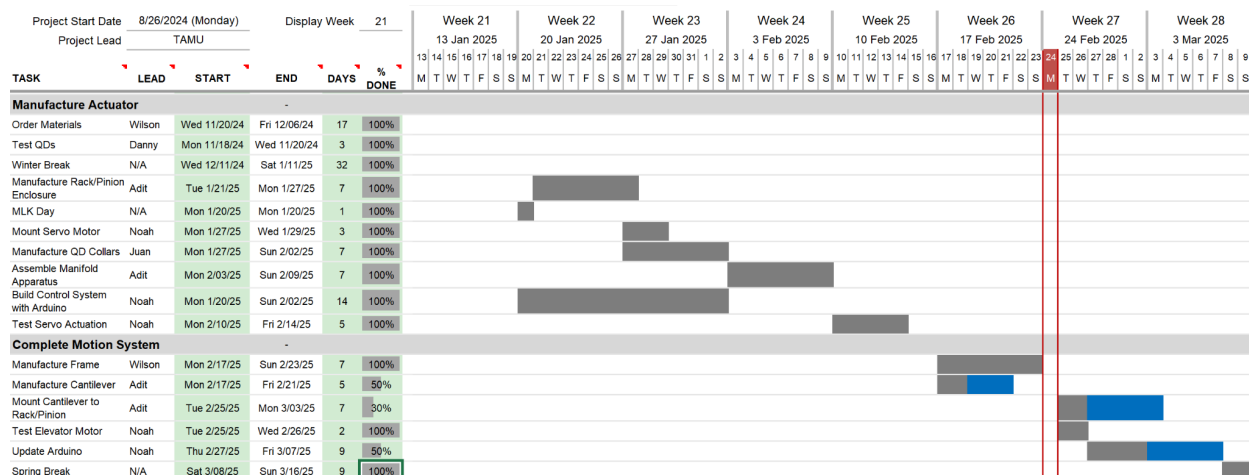


Figure 11: Gantt Chart and Project Plan

5.3 Technology Development Required and Future Work

1. Integration of Control System with Existing Rack Dashboard: The control system needs to be integrated with the existing rack dashboard to allow for seamless user operation and monitoring. System inputs and outputs should be mapped to the current dashboard interface while maintaining ergonomic accessibility and minimizing any additional complexity for operators.
2. Extension of Elevator to Reach All Rack Levels: The elevator mechanism should be extended to reach all rack levels, ensuring full accessibility from top to bottom. The prototype replicates all the dimensions of the existing rack apart from the 42U height.
3. Secure Mounting of the System to the Rack: The system needs to be securely mounted to the rack to ensure stability during operation. Custom mounting brackets and fasteners should be developed to minimize vibration, prevent shifting under load, and maintain precise alignment over time.
4. Addition of Collars to Existing Quick Disconnects (QDs): Collars should be added to the existing quick disconnects (QDs) to increase the surface area that the actuator interacts with. The prototype uses shaft collars to achieve this goal but this is not a very reliable solution and has a number of disadvantages. Incorporating this collar in the design of the QD itself would solve most issues.
5. Reinforcement of Design for Longer Lifespan: The design should be reinforced to extend the system's operational lifespan. Stronger materials, improved structural support, and better stress distribution strategies are recommended to enhance durability and reduce maintenance needs over time.

6. Improvement of Assembly and Maintenance Processes: The design should be improved to make assembly and maintenance easier. Simplifying component interfaces, using standard fasteners, and designing for better service access will reduce installation time, minimize errors, and facilitate quicker repairs and upgrades in the future. Some issues the team recognized that require redesign are the motor housing, due to difficulty in assembly, and the cantilever beam, which is difficult to maintain or replace.

5.4 Limitations of Solution

While the proposed design for the integration of the robotic tool and the rack system offers significant advantages in terms of efficiency and functionality, several limitations and considerations must be addressed to ensure seamless implementation and long-term reliability.

1. Limited Compatibility: The robotic tool system and its associated components may not be universally compatible with all Dell server models or rack configurations. Different server models may have variations in their design, such as rack spacing, connection types, and internal configurations. This lack of uniformity can lead to challenges when deploying the system across a diverse set of server environments. It is critical to assess the specific models and configurations in use within the facility before implementing the system. Compatibility testing and potential customization may be required for certain configurations to ensure optimal performance.
2. Installation Complexity: The installation of the robotic tool and its integration with the existing rack system might require a higher degree of technical expertise than initially anticipated. The initial setup may involve manual adjustments, configuration of the mechanical components, and calibration of the control systems. In addition, there may be requirements for coordination with other parts of the server infrastructure, such as power supply adjustments, wiring, and network integration. For a smooth deployment, thorough training and preparation are essential for the installation team to minimize disruptions and ensure the system is operational in a timely manner.
3. Mechanical Wear and Preventive Maintenance: As with any mechanical system, the robotic tool will experience some level of mechanical wear over time. Moving parts and actuators, in particular, are prone to stress, which can affect the system's performance and longevity. It is important to incorporate the robotic tool into existing preventive maintenance schedules to monitor and address wear and tear proactively. Regular inspections, lubrication, and recalibration may be required to keep the system running smoothly. An effective maintenance plan should be developed to ensure that downtime is minimized and that the robotic tool continues to function efficiently throughout its lifespan.

6. Acknowledgements

Please provide an Acknowledgement paragraph in your final report. Broadly, you should acknowledge all those who helped you by going beyond their normal call of duty, especially those whose help proved crucial to your work or who provided expertise that you lacked. Such people may include individuals from your sponsor, some of your peers, and your mentors (research supervisors or guides), etc

7. References

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2. “What Is Direct Liquid Cooling System? | 2CRSi.” 2CRSI SA, 2021, 2crsi.com/direct-liquid-cooling#:~:text=The%20principle%20of%20the%20DLC,is%20to%20dissipate%20the%20heat.
3. “Dell Technologies Direct Liquid Cooling Support for New PowerEdge Servers | Dell Technologies Info Hub.” Delltechnologies.com, 2023,

8. Appendix

Table 8: Individual Item Cost Breakdown

Description	Category	Unit Price	Quantity	Shipping	Item Total
Elevator Motor and Slider	Consumables	\$114.99	1	\$70.35	\$185.34
Slotted Plate	Consumables	\$6.29	4	\$16.01	\$41.17
Screws	Consumables	\$0.89	4	\$16.01	\$19.57
Hex Nuts	Consumables	\$1.09	4	\$16.01	\$20.37
Arduino	Consumables	\$27.60	1	\$0.00	\$27.60
Raw Aluminum	Consumables	\$28.99	1	\$0.00	\$28.99
External Hose	Consumables	\$35.52	3	\$6.99	\$113.55
Manifold	Consumables	\$103.85	2	\$0.00	\$207.70

Hose Adapter	Consumables	\$4.45	10	\$10.98	\$55.48
Manifold Plug	Consumables	\$1.69	5	\$0.00	\$8.45
Manifold Hoses	Consumables	\$38.93	3	\$0.00	\$116.79
Pinion Gear	Consumables	\$14.14	2	\$0.00	\$28.28
Rack	Consumables	\$115.00	1	\$0.00	\$115.00
Servo Motor for Rack-Pinion	Consumables	\$28.99	2	\$0.00	\$57.98
Step Motor Connection	Consumables	\$10.89	1	\$0.00	\$10.89
Force Tester	Consumables	\$39.99	1	\$0.00	\$39.99
D-Shaft	Consumables	\$2.09	2	\$5.00	\$9.18
Shaft Coupler	Consumables	\$9.99	2	\$5.00	\$24.98
M5 Nuts	Consumables	\$3.78	1	\$0.00	\$3.78
M5 Bolts	Consumables	\$4.42	1	\$0.00	\$4.42
M4 Bolts	Consumables	\$13.24	1	\$0.00	\$13.24
M4 Nuts	Consumables	\$3.11	1	\$0.00	\$3.11
Lead Screw System w/ Motor	Consumables	\$147.99	1	\$70.00	\$217.99
8020 Extrusion Framing	Consumables	\$23.51	3	\$0.00	\$70.53
8020 Extrusion Framing Corner Gussets	Consumables	\$5.75	45	\$0.00	\$258.75
8020 Extrusion Framing Hardware	Consumables	\$2.44	5	\$0.00	\$12.20
Shaft Coupler	Consumables	\$9.99	2	\$4.52	\$19.98
QD Collar 7/8	Consumables	\$7.65	1	\$0.00	\$7.65
Sheet Metal (2 pack)	Consumables	\$29.99	1	\$0.00	\$29.99
QD Collar 15/16	Consumables	\$12.33	1	\$10.98	\$29.99
Arduino Helper Components	Consumables	\$29.99	1	\$0.00	29.99
Limit Switches	Consumables	\$5.99	1	\$0.00	5.99
Barrel Connectors	Consumables	\$3.97	1	\$0.00	3.97
Power Supply	Consumables	\$9.99	1	\$0.00	9.99

QD Collar 1 in	Consumables	\$8.32	1	\$0.00	\$8.32
QD Collar 1 1/8 in	Consumables	\$9.44	1	\$0.00	\$9.44
QD Collar 1 3/16 in	Consumables	\$7.76	1	\$0.00	\$7.76
QD Collar 1 1/4 in	Consumables	\$10.38	1	\$0.00	\$10.38
2 piece collar 15/16	Consumables	\$12.99	3	\$0.00	\$38.97
QD hoses	Consumables	\$34.40	5	\$0.00	\$172.00
ESP32 Boards	Consumables	\$13.99	1	\$0.00	\$13.99
LCD Screens	Consumables	\$11.99	1	\$0.00	\$11.99
Push Buttons	Consumables	\$12.49	1	\$0.00	\$12.49
10 ft Garden Hose 3/4" fittings	Consumables	\$19.99	2	\$0.00	\$39.98
3/4 Shutoff Valve	Consumables	\$24.99	1	\$0.00	\$24.99
Submersible Pump	Consumables	\$59.73	1	\$0.00	\$59.73
Bucket w/ Lid	Consumables	\$6.99	1	\$0.00	\$6.99
Flow Meter	Consumables	\$24.99	1	\$0.00	\$24.99
Control Valve	Consumables	\$65.99	1	\$0.00	\$65.99
3/4 Plugs	Consumables	\$2.22	4	\$0.00	\$8.88
NPT M/F elbow	Consumables	\$12.99	4	\$0.00	\$51.96
FGHT to MNPT	Consumables	\$6.99	6	\$0.00	\$41.94
MNPT to MGHT	Consumables	\$6.99	2	\$0.00	\$13.98
5v buck converter	Consumables	\$14.99	1	\$0.00	14.99
45Kg Servo	Consumables	\$33.99	1	\$0.00	33.99
JST Connectors	Consumables	\$9.99	1	\$0.00	9.99
Screw Connector	Consumables	\$6.99	1	\$0.00	6.99